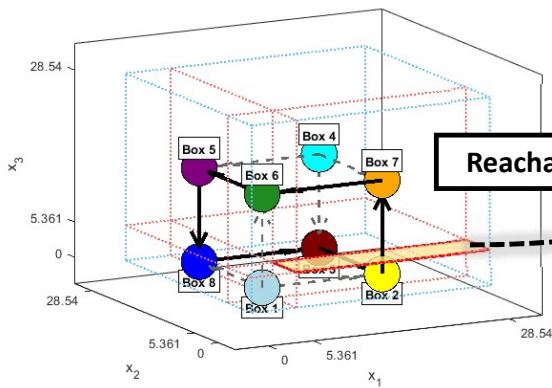


Existence, Localization, and Stability of a Limit Cycle in a Class of Benchmark Biomolecular Oscillators

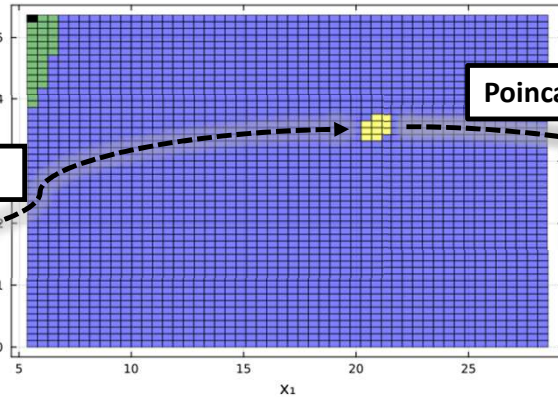
Existence



Reachability Analysis

Construct a positively invariant hypercube and exclude the steady-state point and its stable manifolds. **Brouwer's Fixed Point theorem** applied to the cross-section that maps onto itself guarantees a limit cycle.

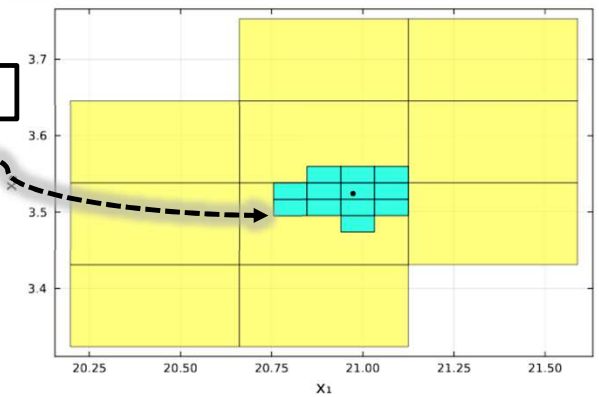
Localization



Poincaré Map

Interval-based Reachability Analysis propagates sets of initial conditions using the system dynamics. Regions that fail to return to the initial set are eliminated, yielding a localized region containing the limit cycle.

Stability



Interval-based Poincaré Map is constructed over the localized region. Stability is certified where the Jacobian of the Poincaré map has eigenvalues less than 1, ensuring contraction toward the limit cycle.

These results add a simple and elegant approach to demonstrating the existence, localization, and stability of limit cycles in biomolecular systems.